

Wang’s Three Laws: Spectral Signals Inside Large Language Model Attention Weights

Fei-Yun Wang*

Independent Researcher, Beijing, China

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Reasoning capability in large language models (LLMs) is typically assessed through benchmark evaluation, which is costly, prompt-sensitive, and requires full inference. We ask whether reasoning capacity can be inferred directly from static attention projection weights, without running any inference. By applying singular value decomposition (SVD) to the Query (Q) and Key (K) weight matrices across five open-source LLM families, we identify two universal cross-model regularities and one precision-depth constraint, which we formalize as *Wang’s Three Laws*. The First Law (Spectral Linear Alignment) states that the Pearson correlation between Q and K singular-value spectra converges toward a theoretical constant $r_{\text{Wang}} = 1$, observed in practice at 0.94–0.98. The Second Law (Spectral Shape Fidelity) states that the normalized spectral mismatch (SSR) decreases systematically in deep layers and is further reduced by reinforcement learning (RL) fine-tuning, with RL-tuned models showing up to 3.92% lower deep-layer SSR than their base counterparts. The Third Law (Precision-Depth-Logic Criterion) derives an analytic upper bound on trainable depth as the minimum of an information-decay limit, a quantization-noise limit, and a dynamic-range limit, explaining why models deeper than 40–80 layers universally adopt BF16 or mixed precision. All findings are reproducible from publicly available model weights using the scripts provided at <https://github.com/emis-framework/math-under-llm>.

I. INTRODUCTION

A. Motivation

Benchmark-based evaluation of LLM reasoning is expensive and prompt-sensitive [1]. Running a full evaluation suite on a single checkpoint may require thousands of GPU-hours and yields results that can be gamed by prompt engineering or benchmark contamination. We ask a different question: *can reasoning capacity be inferred directly from model weights, without running any inference?*

We focus on the attention projection matrices W_Q and W_K and their singular-value spectra. Our central finding is that two universal geometric regularities appear across every model family we examine, and that these regularities correlate monotonically with independently measured reasoning quality.

B. Contributions

1. Formalization of **Wang’s Three Laws**: two empirical spectral regularities and one analytic precision-depth bound.
2. Reproducible verification scripts across four LLM families (Qwen, DeepSeek, LLaMA, Gemma).
3. Quantitative evidence that RL fine-tuning systematically improves deep-layer spectral alignment.

4. Practical guidance for training monitoring, checkpoint selection, and precision planning.

II. BACKGROUND

A. Transformer Self-Attention

The scaled dot-product attention is defined as

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d_h}}\right)V, \quad (1)$$

where the projections are $Q = XW_Q$, $K = XW_K$, $V = XW_V$, and d_h is the head dimension.

B. Singular Value Decomposition

For a weight matrix $M \in \mathbb{R}^{m \times n}$, the SVD is

$$M = U\Sigma V^\top, \quad \Sigma = \text{diag}(\sigma_1, \dots, \sigma_r), \quad \sigma_1 \geq \dots \geq \sigma_r \geq 0. \quad (2)$$

All spectral metrics in this paper are computed on W_Q and W_K extracted from each attention head.

III. WANG’S THREE LAWS

A. First Law — Spectral Linear Alignment

Statement. The Pearson correlation between the Q and K singular-value spectra converges toward a theoretical constant:

$$r(s_q, s_k) \rightarrow r_{\text{Wang}} = 1. \quad (3)$$

* wehelpwe@163.com

Wang’s Constant is defined as the theoretical extreme $r_{\text{Wang}} = 1$; in practice we observe 0.94–0.99.

Interpretation. High Pearson correlation indicates stable spectral alignment between the query and key subspaces, which we hypothesize supports stable information propagation through deep layers. Significant deviations below this range may signal training anomalies or reduced reasoning fidelity.

Empirical evidence is summarized in Table I.

TABLE I. Median and mean Pearson correlation $r(s_q, s_k)$ and SSR across all layers and heads for each model. Models are ordered by reasoning tier (weakest to strongest). The monotone trend—stronger reasoning models exhibit higher r and lower SSR—is consistent with Eq. (3).

Model	Med. r	Mean r	Med. SSR	Mean SSR	Layers
Gemma-4-E2B	0.9183	0.9242	0.015702	0.013537	35
Gemma-4-E4B	0.9585	0.9411	0.009747	0.010080	42
LLaMA-3-8B	0.9813	0.9737	0.006196	0.007009	32
Qwen2.5-14B	0.9795	0.9710	0.006077	0.006710	48
DeepSeek-R1	0.9800	0.9714	0.005948	0.006585	48

The trend is monotone: stronger reasoning models exhibit higher r and lower SSR, consistent with the law’s prediction.

B. Second Law — Spectral Shape Fidelity

Statement. The normalized spectral mismatch between Q and K, defined as

$$\text{SSR} = \frac{1}{d_h} \sum_{i=1}^{d_h} |\tilde{s}_{q,i} - \tilde{s}_{k,i}|, \quad \tilde{s} = \frac{s}{\|s\|_2}, \quad (4)$$

decreases systematically in deep layers. The theoretical extreme is $\text{SSR}_{\text{Wang}} = 0$.

RL amplifies this effect. Reinforcement-learning-tuned models show lower SSR in deep layers than their base counterparts, and the gap grows with depth. Table II compares Qwen2.5-14B (base) and DeepSeek-R1 (RL-tuned) across four layer groups.

TABLE II. Layer-group mean SSR comparison between Qwen2.5-14B (base) and DeepSeek-R1 (RL-tuned), each averaged over 12 consecutive layers. The improvement grows monotonically with depth, reaching 3.92% in layers 36–47, indicating that RL fine-tuning structurally reshapes deep-layer Q/K spectral alignment rather than merely altering output distributions.

Layer group	Qwen2.5-14B SSR	DeepSeek-R1 SSR	Improvement
0–11	0.006852	0.006818	+0.48%
12–23	0.006414	0.006338	+1.17%
24–35	0.006831	0.006704	+1.87%
36–47	0.006743	0.006479	+3.92%

Interpretation. SSR is a quantitative, inference-free proxy for reasoning fidelity. The depth-dependent improvement induced by RL suggests that RL fine-tuning does not merely alter output distributions; it structurally reshapes the geometric relationship between Query and Key attention subspaces.

C. Third Law — Precision-Depth-Logic Criterion

Statement. The maximum trainable depth L_{max} is constrained by three independent limits:

$$L_{\text{max}} = \min(L_{\text{info}}, L_{\text{quant}}, L_{\text{dyn}}), \quad (5)$$

where

$$L_{\text{info}} = 1/\overline{\text{SSR}}, \quad (6)$$

$$L_{\text{quant}} = 3 \cdot 2^{2m} \quad (m = \text{mantissa bits}), \quad (7)$$

$$L_{\text{dyn}} = \log_{\kappa}(\text{MaxFinite}). \quad (8)$$

Example values for standard floating-point formats are given in Table III.

TABLE III. Dynamic-range depth limit L_{dyn} for standard floating-point formats, assuming a representative condition number $\kappa \approx 5$.

Format	Mantissa bits m	MaxFinite	L_{dyn}
FP16	10	6.55×10^4	16
BF16	7	3.39×10^{38}	128

Interpretation. The Third Law explains why models deeper than 40–80 layers universally adopt BF16 or mixed precision: FP16 exhausts its dynamic range at $L_{\text{dyn}} \approx 16$, while BF16 extends this limit to ≈ 128 . The information-decay limit L_{info} provides a complementary SSR-based ceiling that is independent of format.

IV. METHODOLOGY

Models. We analyze five open-source models spanning four architecture families: LLaMA-3-8B [4], Qwen2.5-14B [5], DeepSeek-R1 [3], Gemma-4-E2B, and Gemma-4-E4B [6].

Procedure.

1. Load weight tensors W_Q, W_K for each head and layer.
2. Compute SVD per Eq. (2).
3. Compute Pearson $r(s_q, s_k)$ and SSR per Eqs. (3) and (4).
4. Aggregate across heads; record layer-wise and model-wide statistics.

No inference, tokenization, or prompt evaluation is required. All computation runs on CPU in under five minutes per model.

Verification scripts. `check-gemma.py`, `check-qwen.py`, `check-llama.py`, `check_r1_full.py`, and `qwen-vs-deepseek-all-layers.py` are available at <https://github.com/emis-framework/math-under-llm>.

V. PRACTICAL APPLICATIONS

The three laws suggest several immediately actionable uses:

- **Benchmark-free reasoning assessment.** SSR and r can be computed from any checkpoint without inference, enabling fast model screening.
- **Checkpoint selection.** During training, the checkpoint with the lowest deep-layer SSR is predicted to yield the best reasoning performance.
- **RL progress monitoring.** SSR provides a layer-wise signal of RL-induced structural improvement, independent of reward hacking.
- **Precision planning.** The Third Law gives an analytic guide for choosing FP16 vs. BF16 based on target depth.
- **SSR-regularized fine-tuning.** Adding $\lambda \sum_l |\text{SSR}_l^{\text{pre}} - \text{SSR}_l|$ to the fine-tuning loss anchors the spectral structure that encodes reasoning capability.

VI. FUTURE DIRECTIONS

Several open questions remain:

1. **Validation on closed-source models.** We invite researchers with access to GPT-4/4.5, Claude 3, and Gemini weights to run the five-line verification script and report anonymized SSR profiles.
2. **Theoretical foundations.** Why does gradient descent drive $r \rightarrow 1$? Can the SSR lower bound

be derived from first principles? What is the connection to neural tangent kernels?

3. **Hilbert-space adjoint hypothesis.** If $W_Q \approx f(W_K^\dagger)$ for a spectral-preserving f , Q may be derivable from K analytically, bypassing gradient descent for attention layers entirely.
4. **Spectral surgery.** Can reasoning capability be improved by directly modifying singular values of underperforming layers, without any labeled training data?
5. **MoE and multi-modal extensions.** Do the laws hold per expert in mixture-of-experts models? Does cross-modal attention exhibit different SSR behavior?

VII. CONCLUSION

We have presented Wang’s Three Laws: two universal empirical regularities in the Q/K spectral structure of transformer attention weights, and one analytic precision-depth bound. The First Law establishes that Q and K singular-value spectra are linearly aligned ($r \rightarrow 1$) across all architectures examined. The Second Law shows that normalized spectral mismatch (SSR) decreases with depth and is further reduced by RL fine-tuning. The Third Law provides an analytic ceiling on trainable depth as a function of floating-point precision and dynamic range.

Together, these laws enable *static, inference-free assessment of LLM reasoning capability* from attention weights alone. All results are reproducible from public model weights using the accompanying scripts.

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